Science and Technology Facilities Council
Hartree Centre Cardiff Hub | Canolfan Hartree Hwb Caerdydd

Ariennir gan **Lywodraeth Cymru**
Funded by **Welsh Government**

AI POLLINATION STUDENT COMPETITION 2026

Transforming the Welsh Events Sector Through Responsible AI Innovation

Delivered by the Hartree Centre Cardiff Hub (HCCH) at Cardiff University
Funded by the Event Wales AI Pollination Scheme (Welsh Government)

1. An Opportunity for Students

Are you ready to apply your technical, analytical, or business skills to real-world challenges facing Wales' vibrant events sector? The **Hartree Centre Cardiff Hub (HCCH)** invites students to participate in the **AI Pollination Student Competition**. This initiative challenges individuals or small teams of up to 3 students to solve real world digital transformation and AI adoption hurdles for the events sector in Wales.

Why Enter? Key Competition Rewards:

- **£1,500 Cash / Voucher Total Prize Fund:** Prizes awarded to winning submissions based on technical and strategic merit.
- **Mentoring from AI Experts:** Work with HCCH data scientists during dedicated support drop-in sessions and get your questions answered.
- **High-Profile Industry Showcase:** Winners earn an exclusive invitation to present their solution live at a Showcase Event in March 2027 to event sector companies and Welsh Government.

2. Eligibility & Team Structure

- **Target Audience:** Open to all MSc Taught Students and 3rd Year Undergraduate Students within the School of Computational and Mathematical Sciences.
- **Team Format:** You may enter as an individual or in small teams of up to 3 students. (If a team wins, the prize voucher fund will be shared equally among team members).

3. The Challenges

Choose 1 of the following 3 challenges for your submission. Only 1 submission, accepted per person or team.

◇ Technical Challenge 1 – Knowledge Management

Scenario – A festival logistics company is struggling to keep on top of paperwork. They receive regular reports containing plain text, figures, and tables. Although the reports are well structured, information can be lost as there is no easy way to query the document or condense it down into its key points. The company needs a repository that accepts file uploads and allows them to query and summarise uploaded files. As the company receives paperwork of all kinds of content and structure, the repository should not make any assumptions about file content or layout.

Task – Create an application or pipeline that accepts .docx and .pdf files as input and provides a way to query or summarise the file(s) main points. Your solution must work with a single document that contains text, figures, and tables. Responses to queries should be traceable back to where in the document they came from (e.g. page, section, or table reference). You should aim to address as many of the following as you can:

- Query and summarise plain text content across one or more uploaded documents.
- Extract and answer questions about structured data in tables (e.g. totals, comparisons, specific values).
- Interpret and answer questions about figures and charts (e.g. trends, key values, labelled data points, summarise images).
- Achieve all three within the same document summarisation / query

Data – You will be provided with a selection of files to test your solution with. These will not necessarily be festival / logistics related.

◇ Technical Challenge 2 – Travel Organizer

Scenario – A company is setting up an event that multiple schools want to attend. The number of attendants from different schools can vary significantly, depending on how popular the event is at that school. That creates a challenge for some of those schools when renting bus services to transport the attendants, as they may have too few to justify renting an entire bus. If there are multiple schools in the same area, or on the same route to the event, then those schools would like to share buses between them, which would allow them to split the costs.

Task – Automatically find the best combination of which schools should share buses to minimise total costs. Attendants from individual schools cannot be split between more than one bus, as there may only be a single adult supervisor. As for the cost, choose to implement one or more of the following, in increasing order of complexity:

- Fixed cost per bus rented, depending on bus size.
- Distance between points as the crow flies times bus-size dependent cost.
- Travel time times bus-size dependent cost.

Data – You will be provided with bus costs for each of the options above. You will also be provided with the coordinates of the event, as well as a list of coordinates of places to pick up attendants from, and the number of attendants for each location.

Business Challenge – Responsible AI Implementation Strategy

Scenario – An events company based in South Wales wants to understand how artificial intelligence (AI) could be used to improve its operations. They help clients organise a mixture of sports and cultural events throughout Wales including festivals, marathons, concerts and conferences. They are considering the use of AI across all aspects of the business including for their communications, marketing, improving productivity, planning events for clients and analysing their data. The company recognises the benefits of AI but want to adopt it in a way that considers the following principles:

- Transparency
- Privacy
- Accountability
- Safety
- Security
- Fairness

The company wants to develop a “Responsible AI Implementation Strategy” that enhances the business through the effective and ethical use of AI.

Task – Act as a consultant and develop a “Responsible AI Implementation Strategy” for the company.

The strategy should include:

- Recommendations for where the business could use AI to deliver value and impact
- Consideration of the risks and opportunities associated with AI adoption
- Practical guidance for implementing Responsible AI principles: Transparency, Privacy, Accountability, Safety, Security and Fairness
- Recommendations for governance and policies to support the use of Responsible AI
- Methods for measuring the success and impact of AI adoption

In addition to these requirements, you are encouraged to include any other relevant content or ideas in your strategy. Your submission should be in the form of a report or presentation for senior management. If submitting a presentation, it should be understandable without it being presented live with speaker notes being optional.

Data – You will be provided with a selection of business documents about the company. In addition to these documents, you are encouraged to draw on research, industry standards or other relevant material when developing your strategy.

4. Key Dates & Submission Timeline

Timeline	Date & Deadline	Details
Competition Launch Event	2nd October 11am	Come and find out more information about the competition.
Phase 1: EOI Submission	Wed 7th October 2026 (by 5:00 PM)	Submit a short Expression of Interest (max 500 words) via this form outlining your chosen challenge and proposed idea.
Phase 2: Shortlist Announced	Wed 14th October 2026	Shortlisted entries notified and invited to progress into the Final Development phase.
Phase 3: **Solution Development** **and**		Individually or in your team of 2-3, develop a solution for your chosen challenge. Develop the technical output and prepare a 5-7 min presentation for the panel including the demo of your solution.
HCCH Technical Drop-ins	Tue 20th October and Wed 21st 2026	Optional - participate in dedicated online mentoring & technical advice sessions led by HCCH data scientists.
Phase 4: Final Pitch Event	Wed 28th October 2026	Present your solution in a 5–7-minute presentation (max 10 slides) to an expert judging panel. Winners announced.

5. Assessment & Scoring Rubric

Submissions and final presentations will be evaluated by an expert judging panel (including experts from HCCH, Cardiff Business School, and Event Wales). Each category is scored out of 5 (Total Score: 20 points):

Criteria	Max Score	Excellence Benchmark (Score 5/5)
Communication	5 pts	Clearly articulated solution to the problem statement understandable by a non-expert.
Innovation & AI Approach	5 pts	Highly innovative approach involving novel AI methodologies or creative application of existing algorithms/LLMs to solve previously unsolved problems.
Sector Impact & Outcome	5 pts	Clear identification of beneficiaries (event SMEs, audiences), co-production evidence, and measurable productivity/experience gains for Wales.
Strategic Alignment & Responsible AI	5 pts	Direct alignment with the National Events Strategy for Wales 2022-2030 and strong evidence of inclusive, ethical, and responsible AI governance.

Ready to Enter? Submission Instructions:

1. Prepare your Expression of Interest (EOI): Draft a short proposal of no more than 500 words specifying your team details, selected challenge, proposed AI approach, and expected impact.

3. Complete Submission: Submit your EOI via the following form before 5:00 PM on Wednesday 7th October 2026.

4. Responsible AI Disclosure: If you use generative AI tools to assist in drafting any part of your submission or final presentation, please explicitly declare which tools were used and how.

4. Questions & Support: Contact the HCCH team at Hartree@cardiff.ac.uk for any technical or organizational queries.